Lab 12 - Decision Tree Classifier from Scratch

	Weather	Windy	Play Tennis
S1	Sunny	No	No
S2	Sunny	Yes	No
S3	Overcast	No	Yes
S4	Rain	No	Yes
S5	Rain	Yes	No
S6	Rain	No	Yes
S7	Overcast	Yes	Yes
S8	Sunny	No	No

No.(No) = 4

No(Yes) = 4

T = 8

No → No 2, Yes 3

Yes → No 2, Yes - 1

$\log_2 P_{Yes}$

Yes - 1

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* Step 1 Compute root Entropy

yes = 4 No = 4 , $P(Yes) = \frac{4}{8} = 0.5$ $P(No) = \frac{4}{8} = 0.5$

H (Entropy) — No of Classes

$$H_{root} = \sum_{i=1}^{2} -P_i \log_2 P_i$$

$$= (-P_{yes} \log_2 P_{yes}) + (-P_{No} \log_2 P_{No})$$

$$= \cancel{-0.5}\ 0.5 + 0.5 = 1$$

$$\boxed{H_{root} = 1}$$

$$\log_2(x) = \frac{\log_{10}(x)}{\log_{10}(2)}$$

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* Step 2 Compute Information Gain for all the features = {weather: Sunny, overcast, Rain}

$$H(Sunny) = \sum_{i=1}^{1} -P_i \log_2 P_i = -1 \log_2 1$$

Sunny = (No, No, No) $\frac{3}{3} = 1$

$H_{sunny} = 0$

$H_{overcast}$ = (Yes) Yes) = $H_{overcast} = 0$

H(Rain) $P(Yes) = \frac{2}{3} = 0.66$ $P(No) = 0.33$

$$H_{rain} = -0.66 \log_2(0.66) + (-0.33 \log_2(0.33))$$

$$= 0.912$$

Compute weighted Entropy

$$= \sum \frac{\text{No of Samples}}{\text{Total Samples}} H_{child}$$

$$= \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8} 0.912$$

$$\boxed{= 0.344}$$

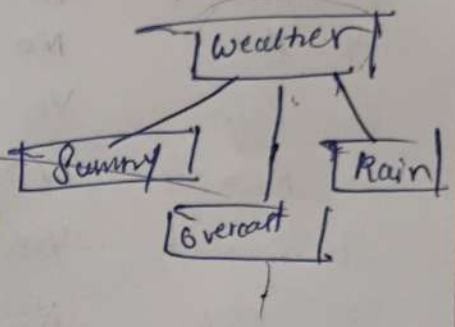
IG Calculation = Hparent − Hchild

= 1 − 0.344

$\boxed{\text{IG weathe} = \underline{0.656}}$

IG (weather) = 0.656

IG (windy) = 0.046



for feature windy

$H(No) \rightarrow P(Yes) = \frac{3}{4} \quad P(No) \ \frac{1}{4}$

$= \sum_{i=1}^{2} -P_i \log_2(P_i)$

$= \left(-0.75(\log_2(0.75)) + (-0.25 \times \log_2(0.2))\right)$

$= \underline{0.81127}$

$H(Yes) \rightarrow P(No) = \frac{2}{5} = 0.4$, $P(Yes) = \frac{3}{5} = 0.6$

$= \sum_{i=1}^{2} -P_i \log_2(P_i)$

$= (-0.4 \log_2(0.4)) + (-0.6 \log_2(0.6))$

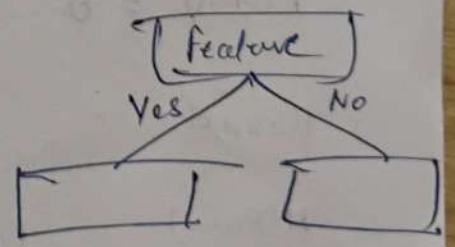
$= \underline{0.970}$

$H(Yes) \rightarrow P(No) = \frac{2}{3} = 0.66$, $P(Yes) = \frac{1}{3} = 0.33$

$= (-0.66 \log(0.66)) + (-0.33 \log(0.33))$

0.923

$\frac{5}{8}$



Weighted H for (windy)

$= \frac{5}{8} \times 0.970 + \frac{3}{8} \times 0.923$

$= 0.9523$

IG $= 1 - 0.9523 = \underline{0.0476}$